INFO 4220 – Networks II: Market Design Spring 2024 – Problem Set 3

Solution

Problem 1: Probabilistic Assignments (10 points)

In a world where cities frequently face challenges ranging from alien invasions to supervillain schemes, a council of superheroes has been formed to protect humanity. This council includes three renowned heroes, each with their own unique powers and city preferences:

Captain Fantastic (CF)	Ms. Mystic (MM)	The Speedster (TS)
Silicon City	Silicon City	Metroville
Metroville	Greenburg	Silicon City
Greenburg	Metroville	Greenburg

Given these preferences, the council decides to use a fair method to determine who goes where.

a. Consider the following assignment. CF gets Silicon City; MM gets Greenburg; TS gets Metroville. (3 points, 1 point each)

   i. Which algorithm could have been used to get this assignment? Propose at least two.

   The specific assignment could have been obtained using the **serial dictatorship algorithm** that uses the following priority order: CC, TS, MM; but also, the priority order CC, MM, TS would have generated the same assignment.

   Or a TTC with initial endowments: CF – Silicon City, MM – Metroville, TS – Greensburg

   Or any other option that creates the same result.

   ii. Is this assignment efficient? Show analytically the steps you take to draw a conclusion.

   Yes, it is efficient as there is no other assignment that would make at least one of the heroes better off without worsening the others.

   iii. Draw the probabilistic assignment matrix associated with this specific assignment.

   Recalling that each deterministic assignment can be represented as a probabilistic assignment, the resulting matrix is:

	Silicon City	Greenburg	Metroville
CF	1	0	0
MM	0	1	0
TS	0	0	1

b. Consider the following two random assignments, β_1 and β_2 described in the matrices below.

$\boldsymbol{\beta_1}$

	Silicon City	Greenburg	Metroville
CF	0.5	0.2	0.3
MM	0.5	0.5	0
TS	0	0.3	0.7

$\boldsymbol{\beta_2}$

	Silicon City	Greenburg	Metroville
CF	0.5	0.3	0.2
MM	0.3	0.5	0.2
TS	0.2	0.5	0.3

Which assignment is stochastically dominant for the three heroes? **Show and exhaustively motivate all the steps you take to draw conclusions.** (2 points – if you don't show your steps but just write the solution, you get zero points).

Solution

Let's consider CF's preference ranking: Silicon City, Metroville, Greenburg. Under the first random assignment:

- probability of getting the most preferred option: 0.5
- probability of getting **at least** the second most preferred option: 0.5+0.3 = 0.8

Under the second random assignment:

- probability of getting the most preferred option: 0.5
- probability of getting **at least** the second most preferred option: 0.5+0.2 = 0.7

We conclude that β_1 is stochastically dominant for CF.

Let's consider now MM's preference ranking: Silicon City, Greenburg and Metroville. Under the first probabilistic assignment:

- probability of getting the most preferred option: 0.5
- probability of getting **at least** the second most preferred option: 0.5+0.5 = 1

Under the second probabilistic assignment:

- probability of getting the most preferred option: 0.3
- probability of getting **at least** the second most preferred option: 0.3+0.5 = 0.8

We conclude that β_1 is stochastically dominant for MM.

Let's consider TS's preference ranking: Metroville, Silicon City and Greenburg. Under the first probabilistic assignment:
- probability of getting the most preferred option: 0.7
- probability of getting **at least** the second most preferred option: 0.7+0 = 0.7

Under the second probabilistic assignment:
- probability of getting the most preferred option: 0.3
- probability of getting **at least** the second most preferred option: 0.3+0.2 = 0.5

We conclude that that β_1 is stochastically dominant for TS as well.

c. Apply the Birkhoff-von Neumann theorem to decompose the random assignment $\boldsymbol{\beta_1}$ in (b). (2 points – show all the steps you need to draw a conclusion otherwise you will be granted zero points).

We want to obtain β_1, namely:

	Silicon City	**Greenburg**	**Metroville**
CF	0.5	0.2	0.3
MM	0.5	0.5	0
TS	0	0.3	0.7

We notice that each of the deterministic assignments in which we will decompose the random assignment must assign a given city to the superhero **for every non-zero probability** in the random assignment matrix. (The reason is because, using linear algebra and under our assumptions, there is no other way to obtain a convex linear combination of '1s' that gives us the probabilities in the table). Therefore, we need deterministic assignments such that:
CF gets Silicon City, Greenburg and Metroville; MM gets Silicon City and Greenburg; TS gets Greenburg and Metroville.

- μ_1**: CF gets Silicon City.** At this point, MM cannot get Silicon City (nor Metroville). Therefore, in this assignment, MM gets Greenburg and TS gets Metroville.

	Silicon City	**Greenburg**	**Metroville**
CF	1	0	0
MM	0	1	0
TS	0	0	1

- μ_2**: CF gets Greenburg**. At this point, MM can only get Silicon City and TS can only get Metroville.

	Silicon City	**Greenburg**	**Metroville**
CF	0	1	0
MM	1	0	0
TS	0	0	1

- μ_3**: CF gets Metroville**. This means that TS can only get Greenburg and, therefore, MM can only get Silicon City.

	Silicon City	**Greenburg**	**Metroville**
CF	0	0	1
MM	1	0	0
TS	0	1	0

To obtain the first probability assignment, we need that:

$$Prob(\mu_1) = 0.5; Prob(\mu_2) = 0.2; Prob(\mu_3) = 0.3.$$

This is the only possible decomposition into deterministic assignments under the assumptions of the model, namely:
1. each individual gets a city with non-zero probability
2. each city gets an individual with non-zero probability.
3. probabilities sum to one.

d. Is $\boldsymbol{\beta_1}$ **ex-post efficient?** Motivate the answer and show all the steps you need to draw a conclusion (1 point).

Let's consider the deterministic assignments. M1 is ex-post efficient because CF and TS are getting their top choice, while the only way for MM to improve the assignment (second-top choice) is to have Silicon City, but this would make CF worse off. M2 is ex-post efficient, as MM and TS are getting their top choices, while the only way for CF to improve her assignment to the least preferred choice is to make either TS or MM worse off. M3 is ex-post efficient, as MM is getting the top choice, CF is getting the second-top choice but cannot improve the assignment without making MM worse off, TS is getting the least preferred choice, but cannot improve the assignment without making both worse off.

e. Use the eating algorithm to find an ex-ante efficient assignment for this problem. (2 points - if you don't show and explain your steps and just write the solution, you will get 0 points)

Time 0: CF and MM start eating Silicon City, while TS starts eating Metroville.

Time ½: CF and MM end eating Silicon City, (1/2 each), while TS has eaten ½ of Metroville, therefore there is ½ left of Metroville. CF starts eating Metroville and MM starts eating Greenburg.

Time ¾: CF had ¼ of Metroville, and TS had ½+1/4 =3/4 of Metroville. Metroville has been completely eaten. MM has eaten ¼ of Greenburg (3/4 left). CF and TS start eating Greenburg as well.

End: CF and TS get ¼ each of Greenburg, while MM got ½ of it. The resulting probabilistic assignment is:

	Silicon City	**Greenburg**	**Metroville**
CF	½	¼	¼
MM	½	½	0
TS	0	¼	¾

Problem 2: Course Assignment (10 points, 2.5 each)

Cristiana, Fengyang, and Jose are three Ph.D. students who need to enroll in two Core classes in Information Science to fulfill their Ph.D. requirements. Networks and Behavior core courses have two slots, while Design and Ethics courses have only one. For this problem we assume that, depending on how much they like the courses, students obtain a utility from taking the course. The utility is indicated in parenthesis in the table with their preferences below. The utilities students obtain are only known to them. For matching purposes, they on ly provide an ordered preference list, which is used to assign students to courses.

Cristiana	**Fengyang**	**Jose**
Behavior (10)	Networks (10)	Ethics (10)
Networks (6)	Behavior (5)	Networks (7)
Ethics (3)	Ethics (3)	Design (4)
Design (1)	Design (2)	Behavior (1)

Please remember to show and explain all your steps.

a. Assuming students report their true preferences, and that course assignment are done only considering students' preferences. What is the expected utility each student obtains if the courses are assigned using a random serial dictatorship (hint: consider all possible random orders in which students can pick?)

There are six possible orders of priority shown in the first column. The other columns show the course allocation and payoffs are in parentheses.

Priority Orders	Fengyang	Cristiana	Jose
F>C>J	N&B (15)	B&N (16)	E&D (14)
F>J>C	N&B (15)	B&D (11)	E&N (17)
C>F>J	N&B (15)	B&N (16)	E&D (14)
C>J>F	B&D (7)	B&N (16)	E&N (17)
J>F>C	N&B (15)	B&D (11)	E&N (17)
J>C>F	B&D (7)	B&N (16)	E&N (17)

0.5 points for each + 0.1 points if everything is good: 1.6 points

$$EU_F = \frac{(15 * 4 + 7 * 2)}{6} = \frac{74}{6} = 12.33$$

$$EU_C = \frac{(16 * 4 + 11 * 2)}{6} = \frac{86}{6} = 14.33$$

$$EU_J = \frac{(17 * 4 + 14 * 2)}{6} = \frac{96}{6} = 16$$

0.3 each: 0.9 points.

b. The Networks course is too challenging to have too many students enrolled, so slots are reduced to 1. What is the expected utility of the students using the same mechanism (and the same assumptions) as in (a)? Who suffers the greatest expected utility loss due to the change of capacity in Networks? Is there any way they (i.e. the one that suffers the greatest drop in expected utility) could misrepresent their priorities and obtain a higher expected utility (assume that everyone else submits true preferences)?

The problem is similar to before, but Networks' class has only one place available:

Priority Orders	Fengyang	Cristiana	Jose
F>C>J	N&B (15)	B&E (13)	D (4)
F>J>C	N&B (15)	B (10)	E&D (14)
C>F>J	B&E (8)	B&N (16)	D (4)
C>J>F	B (5)	B&N (16)	E&D (14)
J>F>C	B&D (7)	B (10)	E&N (17)
J>C>F	B (5)	B&D (11)	E&N (17)

0.3 each: 0.9 points.

$$EU_F = \frac{(15*2+8+7+5*2)}{6} = \frac{55}{6} = 9.17$$

$$EU_C = \frac{(16*2+10*2+13+11)}{6} = \frac{76}{6} = 12.67$$

$$EU_J = \frac{(17*2+14*2+4*2)}{6} = \frac{70}{6} = 11.67$$

0.3 each: 0.9 points.

The student who faces the highest decrease in utility is Jose. In principle, one way to strategically submit preferences (untruthfully) would be to give higher ranking to options that are less popular and give lower ranking to more popular (but more preferred options). Jose's preferences are already ranking options with low popularity already, so there should not be any way for him to strategically submit different preferences. Given that all courses but 'Behavior' have only one slot, submitting untruthful rankings would lead to potentially lower utility. In an assignment problem the random serial dictatorship is strategyproof.

0.3 for explaining that Jose is the one that was the most impacted by the new policy and 0.4 for saying there is no way to submit his preferences in order to improve her utility. Total 0.7 points.

c. Cornell decides to use the HBS draft mechanism. Assuming students submit their true preferences (and capacities are as in part (b)), what is the expected utility each student obtains? Who would prefer this mechanism? Could Cristiana improve her utility by changing her preferences (assuming Fengyang and Jose submit their true preferences)?

Remember that one course is picked at a time and: (odd steps) we follow the ranking while (even steps) we follow the reverse. For instance, if the priority order is F>C>J, then in step 1 Fengyang is the first one to choose, followed by Cristiana and Jose, while in step 2, Jose is the first one to choose. The resulting allocation is:

Priority Orders	Fengyang	Cristiana	Jose
F>C>J	N&B (15)	B (10)	E&D (14)
F>J>C	N (10)	B&D (11)	E&B (11)
C>F>J	N&B (15)	B (10)	E&D (14)
C>J>F	N&B (15)	B (10)	E&D (14)
J>F>C	N&B (15)	B&D (11)	E (10)
J>C>F	N&B (15)	B&D (11)	E (10)

0.2 each: 0.6 in total.

$$EU_F = \frac{(15*5+10)}{6} = \frac{85}{6} = 14.17$$

$$EU_C = \frac{(10*3+11*3)}{6} = \frac{63}{6} = 10.5$$

$$EU_J = \frac{(14 * 3 + 10 * 2 + 11)}{6} = \frac{73}{6} = 12.17$$

0.2 each: 0.6 in total.

Cristiana might strategically submit her preferences as follows: N, B, E, D. In this way, she will get Networks every time she is the first one to choose, or the second one after Jose. Namely in those cases (in blue below)

Priority Orders	Fengyang	Cristiana	Jose
F>C>J	NB (15)	B (10)	ED (14)
F>J>C	N (10)	BD (11)	EB (11)
C>F>J	B (5)	NB (16)	ED (14)
C>J>F	BD (7)	N(6)	ED (14)
J>F>C	NB (15)	BD (11)	E (10)
J>C>F	BD (7)	NB (16)	E (10)

1 for showing an alternative preference submission for Cristiana which increases utility.

$$EU_C^* = \frac{(10 + 11 * 2 + 16 * 2 + 6)}{6} = \frac{80}{6} = 11.67$$

0.3 for calculating the new utility.

d. A new mechanism is proposed. In step 1, students are assigned the course they list as their first preferences. If a course has fewer open seats than students interested, who gets to enroll is determined randomly with each student getting the same probability to enroll (those that are not selected don't get any course in this step). In subsequent steps, the process is repeated with the second highest preferences. The process continues until there are no more open seats available, or until all students have enrolled in the maximum number of courses that they can enroll in. Find the course assignment that results from using this algorithm (assume course capacities are as in part (b) and students report their true preferences) Who would prefer this mechanism over the options proposed above? [Clarification: If at step nn, the course that student ii listed in position nn is full, then student ii does not get a course in that step]

Step 1:

Cristiana Behavior, Fengyang Networks, and Jose Ethics.

They all get the courses.

Step 2:

Cristiana →Networks (not available), Fengyang →Behavior, and Jose →Networks (Not available).

Step 3:

Cristiana →Ethics (not available) and Jose →Design.

There are no more slots available in courses.

0.4 each step: 1.2 in total.

$$EU_F = 10 + 5 = 15$$

$$EU_C = 10$$

$$EU_J = 10 + 4 = 14$$

0.3 each utility: 0.9 in total.

Fengyang and Jose prefer this algorithm over HBS and RSD.
0.4 points

Problem 3 (10 points)

Consider a school choice problem as below. Each school has a capacity of 1. The students' preferences and the weak priorities of these schools are given in the following table.

Alice	Bob	Carol	Denis
S_4	S_2	S_2	S_3
S_2	S_1	S_4	S_1
S_1	S_4	S_3	S_2
S_3	S_3	S_1	S_4
S_1	S_2	S_3	S_4
Bob	Alice	Alice	[Carol, Denis]
Carol	[Bob, Carol]	Bob	Alice
Denis	Denis	Carol	Bob
Alice		Denis	

For parts (a) – (d) of this problem assume schools break the ties using a single tie breaking rule, and that the lottery resulted in the order Alice > Bob > Carol > Denis.

Please remember to show and explain all your steps.

a. Use the Immediate Acceptance Algorithm to find a school assignment (assuming students submit their true preferences). Is the assignment efficient? Is it stable? (demonstrate efficiency and stability) (1 point)

Step 1: Alice -> S4, Bob -> S2, ~~Carol -> S2~~, Denis -> S3

Step 2: ~~Carol -> S4,~~ but S4 has filled its capacity so Carol was rejected.

Step 3: ~~Carol -> S3,~~ but S3 has filled its capacity so Carol was rejected.

Step 4: Carol -> S1

Assignment: $\mu(Alice) = S_4, \mu(Bob) = S_2, \mu(Carol) = S_1, \mu(Denis) = S_3$

(0.25 pt for correct steps and 0.25 pt for correct assignment.)

The assignment is efficient since Alice, Bob and Denis have their most preferred schools and don't want to switch**. (0.25pt for correctly explaining efficiency.)**

The assignment is not stable. It is individual rational and non-wasteful, but Carol justifiably envies Alice for S4, and justifiably envies Denis for S3. (Not justified envies Bob for S2 since they have same priority.) **(0.25pt for correctly explaining instability, not granting points if stating Carol justified envies Bob for S2 – it's not justified envy since they have the same priority. Finding just one justified envy is enough, no need to list them all)**

b. If the immediate acceptance strategy algorithm is used. Find the optimal strategy students (consider all students) should use when submitting their preferences (remember, use only the order Alice > Bob > Carol > Denis for this question). (1 point)

Carol would want to put S_4 as her top choice, that will make Alice get rejected at S_4. Alice will want to put S_2 as her top choice, that will make Bob get rejected at S_2. Bob will want to put S_1 as his top choice. With this strategy, everyone ends with their top 2 schools. The submitted preference is as below. (Note that any preference ordering that matches the first row of the table below is equivalent and equally correct)

Alice	Bob	Carol	Denis
S_2	S_1	S_4	S_3
S_4	S_2	S_2	S_1
S_1	S_4	S_3	S_2
S_3	S_3	S_1	S_4

0.25 pt for each correct optimal strategy (x4 = 1). If not specifying strategy for each individual but stating "they should put the school that they have higher order as their top choice", get 0.25 pt.

c. Used the deferred acceptance algorithm to find an assignment. (1 point)

Step 1: Alice -> S4, Bob -> S2, ~~Carol -> S2~~, Denis -> S3

Step 2: Carol -> S4, (~~Alice -> S4~~ from step 1).

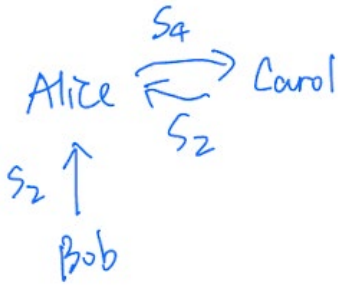
Step 3: Alice -> S2,(~~Bob -> S2~~ from step 1).

Step 4: Bob -> S1

Assignment: $\mu'(Alice) = S_2, \mu'(Bob) = S_1, \mu'(Carol) = S_4, \mu'(Denis) = S_3$

0.5 pt for correct process, 0.5 pt for correct assignment.

d. Now try to improve the assignment you found in (c) using improvement cycles (2 points)



Assignment: $\mu''(Alice) = S_4, \mu''(Bob) = S_1, \mu''(Carol) = S_2, \mu''(Denis) = S_3$

1pt for the correct graph (deduct 0.25pt if not listing Bob), 1pt for correct assignment.

e. The district is deciding whether to use a single or a multiple tie breaking rule. Find the assignments that could result with each of these rules. Write down the probabilistic assignment that results from using each rule (assume they use D.A. without improvements cycles). (4 points)

1) When using single tie breaking, there are 24 possible orders. However, this translate into 4 ways of arranging the priorities at S_2 and S_4

μ_1		μ_2		μ_3		μ_4	
S_2	S_4	S_2	S_4	S_2	S_4	S_2	S_4
Alice	Carol	Alice	Denis	Alice	Carol	Alice	Denis
Bob	Denis	Bob	Carol	Carol	Denis	Carol	Carol
Carol	Alice	Carol	Alice	Bob	Alice	Bob	Alice
Denis	Bob	Denis	Bob	Denis	Bob	Denis	Bob

Running DA with each of those possible orders gives us 2 possible assignments:

$\mu_1(Alice) = S_2$, $\mu_1(Bob) = S_1, \mu_1(Carol) = S_4, \mu_1(Denis) = S_3$

$\mu_2 = \mu_1$

$\mu_3(Alice) = S_4$, $\mu_3(Bob) = S_1, \mu_3(Carol) = S_2, \mu_3(Denis) = S_3$

$\mu_4 = \mu_3$

If we list all possible orders from single tie-breaking, the summary of probability for each possible order combination considering the two ties (in S_2 and S_4), **dependently**, and the resulting assignment are as below (see table in appendix for all possible orders):

S_2	S_4	P(STB)	Assignment from DA
B>C	C>D	4/24	μ_1
B>C	C<D	8/24	μ_1
B<C	C>D	8/24	μ_3
B<C	C<D	4/24	μ_3

Hence, for single tie breaking, the probability for reaching $\mu_1 = 4/24 + 8/24 =$ ½, the probability for reaching $\mu_3 = 4/24 + 8/24 =$ ½.

1pt for correctly showing the probabilistic assignment for STB (orders and probability). 0.5 pt for correctly listing 2 possible assignments, 0.5pt for correctly listing the probability for the 2 assignments. 2pts for STB in total. Deduct points for the process accordingly (0.25 each) if the orders, probability or resulting assignments are incorrect.

Alternative solution for STB: it's ok if student omits Alice when listing the probability for each possible orders (since Alice is not in any tie). There will be 6 possible orders between Bob, Carol and Denis, and the probability for each is 1/6 (as you can see from Appendix). The summary for probability for each combination and resulting assignment is:

STB: BCD orders	S_2	S_4	P(STB)	Assignment from DA
B>C>D	B>C	C>D	1/6	μ_1
B>D>C	B>C	C<D	1/6	μ_1
C>B>D	B<C	C>D	1/6	μ_3
C>D>B	B<C	C>D	1/6	μ_3
D>B>C	B>C	C<D	1/6	μ_1
D>C>B	B<C	C<D	1/6	μ_3

Hence, for single tie breaking, the probability for reaching $\mu_1 = 3/6 =$ ½, the probability for reaching $\mu_3 = 3/6 =$ ½. The probabilistic assignment then is:

	S_1	S_2	S_3	S_4
Alice	0	1/2	0	1/2
Bob	1	0	0	0
Carol	0	1/2	0	1/2
Denis	0	0	1	0

2) When using multiple tie breaking, there are 4 possible combinations for the orders and 2 possible assignments (the assignments are the same as in STB). The summary of probability for each possible order combination considering the two ties (in S_2 and S_4), **independently,** and the resulting assignment are added in the table:

S_2	S_4	P(STB)	P(MTB)	Assignment from DA
B>C	C>D	4/24	½* ½= ¼	μ_1
B>C	C<D	8/24	½* ½= ¼	μ_1
B<C	C>D	8/24	½* ½= ¼	μ_3
B<C	C<D	4/24	½* ½= ¼	μ_3

Hence, for multiple tie-breaking, the probability for reaching $\mu_1 = 1/4 + 1/4 =$ ½, the probability for reaching $\mu_3 = 1/4 + 1/4 =$ ½. (i.e. same as above)

1pt for correctly showing the probabilistic assignment for MTB (orders and probability). 0.5 pt for correctly listing 2 possible assignments, 0.5pt for correctly listing the probability for the 2 assignments. 2pts for MTB in total. Deduct points for the process accordingly (0.25 each) if the orders, probability or resulting assignments are incorrect.

3) In summary, the probabilistic assignment from single tie breaking and multiple tie breaking rule are the same in this problem: there are two possible assignments, each with the probability 1/2.

$\mu_1(Alice) = S_2$, $\mu_1(Bob) = S_1, \mu_1(Carol) = S_4, \mu_1(Denis) = S_3$
$\mu_3(Alice) = S_4$, $\mu_3(Bob) = S_1, \mu_3(Carol) = S_2, \mu_3(Denis) = S_3$

(Note: 2pts should already be granted in previous steps if only showing assignments and their probability. Also, no need to show the exhausted list of the orders as in Appendix, but need to show the probability for combinations of orders and the resulting assignment.)

f. What would you recommend using in this case. Justify your choice. (1 point)

Credits are granted for a reasonable justification.

- In this case it seems that STB was more complicated than MTB and the result is the same. A justification could be based on complexity. (note complexity could be argued both ways, some may find STB was simpler. In this case MTB seems simpler due to the structure of the weak ties, but in general MTB will lead to many more possible outcomes)
- Some could say it doesn't matter because the result is the same.

- Or justification could be based on the result we saw in class, which says that while MTB seems fairer, in practice STB produce better results (it didn't happen in this case, but when there are assignments that could be generated through MTB and cannot be generated through STB, they are not going to be student optimal)
- Other justifications are OK as long as they are justified either by the results of this problem, or the general properties we discussed in class.

Appendix 1. all possible orders from STB (ABCD)

STB orders				B > C	C > D	Orders (BCD)	School 2 priority	School 4 priority	DA without IC results
A	B	C	D	1	1	BCD	ABCD	CDAB	μ_1
A	B	D	C	1	0	BDC	ABCD	DCAB	μ_1
A	C	B	D	0	1	CBD	ACBD	CDAB	μ_3
A	C	D	B	0	1	CDB	ACBD	CDAB	μ_3
A	D	B	C	1	0	DBC	ABCD	DCAB	μ_1
A	D	C	B	0	0	DCB	ACBD	DCAB	μ_3
B	A	C	D	1	1	BCD	ABCD	CDAB	μ_1
B	A	D	C	1	0	BDC	ABCD	DCAB	μ_1
B	C	A	D	1	1	BCD	ABCD	CDAB	μ_1
B	C	D	A	1	1	BCD	ABCD	CDAB	μ_1
B	D	C	B	1	0	BDC	ABCD	DCAB	μ_1
B	D	A	C	1	0	BDC	ABCD	DCAB	μ_1
C	A	B	D	0	1	CBD	ACBD	CDAB	μ_3
C	A	D	B	0	1	CDB	ACBD	CDAB	μ_3
C	B	A	D	0	1	CBD	ACBD	CDAB	μ_3
C	B	D	A	0	1	CBD	ACBD	CDAB	μ_3
C	D	A	B	0	1	CDB	ACBD	CDAB	μ_3
C	D	B	A	0	1	CDB	ACBD	CDAB	μ_3
D	A	B	C	1	0	DBC	ABCD	DCAB	μ_1
D	A	C	B	0	0	DCB	ACBD	DCAB	μ_3
D	B	A	C	1	0	DBC	ABCD	DCAB	μ_1
D	B	C	A	1	0	DBC	ABCD	DCAB	μ_1
D	C	A	B	0	0	DCB	ACBD	DCAB	μ_3
D	C	B	A	0	0	DCB	ACBD	DCAB	μ_3